

Full-production discretization P3: explicit energy-to-H2 ball envelope

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This supplement makes the first previously qualitative input to the P3 positive-time solution-ball route explicit. It applies only to the canonical P1/P2 three-component gradient flow on the fixed 16^3 torus, with positive regularisers and $\eta_{\text{shell}} = 0$. For each declared H^2 radius R it computes an upper energy $E_+(R)$ and an all-time common bound $M_2(R)$:

$$\|u_0\|_{H^2} \leq R \implies \sup_{t \geq 0} \|u(t)\|_{H^2} \leq M_2(R). \quad (1)$$

It does not yet compute the positive-time H^6 bound, the residual/aliasing constant $C(R, \tau, T)$, or a continuum error bar for a Sector-B solver run.

1 Fourier embedding bound

Use the P2-compatible norm

$$\|u\|_{H^2}^2 = |\mathbb{T}^3| \sum_{n \in \mathbb{Z}^3} (1 + |k_n|^4) |\hat{u}_n|^2.$$

By Cauchy–Schwarz,

$$\|u\|_{\infty}^2 \leq \frac{S}{|\mathbb{T}^3|} \|u\|_{H^2}^2, \quad S = \sum_n (1 + |k_n|^4)^{-1}. \quad (2)$$

For a max-norm shell m , there are $24m^2 + 2$ points and $|k_n|^4 \geq (2\pi/L_{\text{max}})^4 m^4$. Summing through M and bounding the tail by the integrals of m^{-2} and m^{-4} gives

$$S \leq 1 + \sum_{m=1}^M \frac{24m^2 + 2}{1 + am^4} + \frac{24}{aM} + \frac{2}{3aM^3}, \quad a = (2\pi/L_{\text{max}})^4. \quad (3)$$

The audit uses $M = 8192$ and obtains $S \leq 445.8565811405$, with tail at most 0.1231917871. Since $|\mathbb{T}^3| = 4096$, (2) yields $\|u\|_{\infty}^2 \leq 0.1088517044 \|u\|_{H^2}^2$.

2 Energy upper and lower bounds

The hash-pinned embedded Pauli generators have operator norm one. Therefore $|J_A| \leq 2|u||\nabla u|$ and $|K_A| \leq 4|u||\nabla u|$. With the P1 coefficients, the absolute Class-II contribution is bounded by

$$(2|a| + 8|b| + 8|c|) \int |u|^2 |\nabla u|^2,$$

whose computed coefficient is 0.04275. Combining this with (2), the linear upper coefficient 2.1566713536, and the quartic/sextic terms gives

$$F(u_0) \leq E_+(R) = \frac{L_+}{2} R^2 + Q_+ R^4 + S_+ R^6. \quad (4)$$

Here $Q_+ = 0.01402826340$ and $S_+ = 0.003199147258$ are calculated from pinned coefficients, not fitted from trajectories.

P2 supplies energy dissipation and

$$F(u) \geq \frac{c_{H^2}}{2} \|u\|_{H^2}^2 - C_Y |\mathbb{T}^3|, \quad c_{H^2} = 0.2048572626782363, \quad C_Y = 0.010098435197886498. \quad (5)$$

Equations (4)–(5) prove (1) with

$$M_2(R)^2 = \frac{2\{E_+(R) + C_Y |\mathbb{T}^3|\}}{c_{H^2}}. \quad (6)$$

3 Executed enclosure

For the declared representative radii, the standalone audit returns:

R	$E_+(R)$	$M_2(R)$
0.5	0.2705106723	20.16098869
1.0	1.0955630875	20.35977321
2.0	4.7425403461	21.21615946

It validates P1 backend, P1 manifest, and P2 manifest hashes; positivity of coercivity, sextic, and regularisers; the generator convention; the series bound; and monotonicity/dominance of (6). All 13 checks pass.

4 Devil’s-advocate and mandatory adversarial review

Tail direction. The shell count and integral tail in (3) are upper bounds, so truncation cannot make S too small. **Class-II factors.** The 2, 8, 8 coefficients follow from the hash-checked unit-norm Pauli convention. **Norm convention.** The fourth-power Fourier weight is the P2 coercivity norm. **Coarseness.** M_2 is deliberately large because the global Young lower bound includes the whole torus volume; it is an enclosure, not a sharp dynamical prediction. **Limit cases.** This does not cover a nonzero shell term, zero floors, infinite volume, or the historical non-variational backend. **No smoothing shortcut.** An H^2 envelope does not prove a computable H^6 envelope. External review of the Class-II inequality and explicit semigroup/Moser constants remains invited.

5 What remains open

The next step must turn $M_2(R)$ into an explicit positive-time $B_6(R, \tau, T)$ using named semigroup and Moser constants. Only then can P3 compute $C(R, \tau, T)$, propagate it through a dealiased finite-time evolution estimate, and consider a tier review.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Explicit all-time H2 envelope M2(R) for declared canonical P2 initial H2 balls.
Scope:	Fixed T3, canonical P1/P2 functional, eta_shell=0.
Dependencies:	A1 production functional; A2 T6 well-posedness.
Evidence grade:	ANALYTIC + EXECUTED; T3 partial stage.
Reproduction command:	python codes/foundations/ a3_full_production_energy_ball_envelope.py
Expected output:	13/13 PASS; A3-FULL-ENERGY-BALL-ENVELOPE-PASS.

Falsification gate: A3-FULL-DISCRETIZATION-CLOSURE.
Tier before / after: T3 / T3.
No-overclaim statement: Not an H6 or C(R,tau,T) enclosure, a solver
continuum certificate, or a T5/T6/T7 result.
Next required action: Explicit positive-time H6 smoothing majorant.